PHL1010: Practice Problems
(03/11/2025)

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1. Problems from Griffiths (3rd edition)
   4.5, 4.6, 4.16, 4.28
   5.6 (b), 5.9, 5.10(b), 5.13, 5.23, 5.25, 5.36

2. In a region, magnetic field **B** points in the x-direction and **E** in the z-direction. A particle at rest and carrying mass m and charge q is released from the origin with velocity (a) $\vec{v}(0) = \frac{E}{B}\hat{y}$, (b) $\vec{v}(0) = \frac{E}{B}(\hat{y} + \hat{z})$. Find and sketch the trajectory of the particle for both the cases.

3. A current I flows down a wire of radius a. If it is uniformly distributed over the surface, what is the surface current density K? If it is distributed in such a way that the volume current density is inversely proportional to the distance from the axis, what is J?

4. A uniformly charged solid sphere, of radius R and total charge Q, is centered at the origin and spinning at a constant angular velocity ω about the z axis. Find the current density **J** at any point (r, θ, φ) within the sphere.

5. Consider two infinite straight line charges λ, which are distance d apart and are moving along at a constant speed v (Fig. 1). How great would v have to be in order for the magnetic attraction to balance the electrical repulsion? Work out the actual number. Is this a reasonable sort of speed?

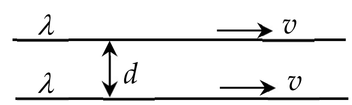

Figure 1

6. A long solid metal cylinder rotates with angular velocity about its axis of symmetry. The cylinder is in a homogeneous magnetic field B parallel to its axis. What is the resultant charge distribution inside the cylinder? Is there an angular velocity for which charge density is everywhere zero?

7. A semicircular wire carries a steady current I (ignore the circuit which connects this semicircular wire to a battery). Define source point and write $\vec{r}\,'$ and then $\vec{\mathscr{r}}$ in Cartesian coordinate system. Find the magnetic field at a field point P on the other semicircle (See Figure 2).

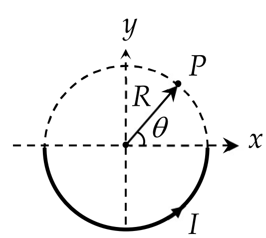

Figure 2